

Namespace ASEAssignmentDemo.Tests

Classes

[AppCanvasTests](#)

[UnitTest1](#)

Class AppCanvasTests

Namespace: [ASEAssignmentDemo.Tests](#)

Assembly: ASEAssignmentDemo.Tests.dll

```
public class AppCanvasTests
```

Inheritance

[object](#)  ← AppCanvasTests

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

Circle_ValidRadius_DrawsCircle()

Tests the AppCanvas.Circle(int, bool) method by verifying that drawing a circle with a valid radius executes without exceptions.

```
[Fact]  
public void Circle_ValidRadius_DrawsCircle()
```

DrawTo_ValidPosition_UpdatesPositionAndDrawsLine()

Tests the [DrawTo\(int, int\)](#) method by verifying that drawing to a valid position correctly updates the pen's position and executes a line-drawing operation.

```
[Fact]  
public void DrawTo_ValidPosition_UpdatesPositionAndDrawsLine()
```

MoveTo_ValidCoordinates_UpdatesPenPosition()

Tests the [MoveTo\(int, int\)](#) method by checking if moving the pen to specified coordinates updates the pen's position correctly.

```
[Fact]
public void MoveTo_ValidCoordinates_UpdatesPenPosition()
```

MultilineProgram_ExecutesMultipleCommands_CorrectlyUpdatesCanvas()

Tests a sequence of drawing commands on the [AppCanvas](#) to verify that multiple operations are correctly executed in sequence, updating the canvas state as expected.

```
[Fact]
public void MultilineProgram_ExecutesMultipleCommands_CorrectlyUpdatesCanvas()
```

Triangle_InvalidDimensions_ThrowsException()

Tests the [Tri\(int, int\)](#) method with invalid dimensions to verify that it throws an appropriate exception.

```
[Fact]
public void Triangle_InvalidDimensions_ThrowsException()
```

Triangle_ValidDimensions_DrawsTriangle()

Tests the [Tri\(int, int\)](#) method by verifying that drawing a triangle with valid width and height executes without exceptions and updates the pen position correctly.

```
[Fact]
public void Triangle_ValidDimensions_DrawsTriangle()
```

Triangle_WithPenColorChange_DrawsTriangle()

Tests the [Tri\(int, int\)](#) method by verifying that changing the pen color and drawing a triangle works as expected.

```
[Fact]
public void Triangle_WithPenColorChange_DrawsTriangle()
```

WriteText_ValidText_DrawsTextOnCanvas()

Tests the [WriteText\(string\)](#) method by verifying that drawing text on the canvas executes without exceptions.

```
[Fact]
public void WriteText_ValidText_DrawsTextOnCanvas()
```

Class UnitTest1

Namespace: [ASEAssignmentDemo.Tests](#)

Assembly: ASEAssignmentDemo.Tests.dll

```
public class UnitTest1
```

Inheritance

[object](#)  ← UnitTest1

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

Test1()

```
[Fact]  
public void Test1()
```

Namespace ASE_Assignment_Demo

Classes

[AppCanvas](#)

Represents a drawable canvas that supports various shapes and text operations. Provides functionality to draw geometric shapes, move the pen, and manage canvas properties.

[Form1](#)

Represents the main form of the application where drawing operations can be performed. This form integrates drawing functionality using the BOOSE library, allowing users to execute commands and display graphical output.

Interfaces

[IAppCanvas](#)

Extends the base interface with additional functionalities for canvas operations.

Class AppCanvas

Namespace: [ASE Assignment Demo](#)

Assembly: ASE Assignment Demo.dll

Represents a drawable canvas that supports various shapes and text operations. Provides functionality to draw geometric shapes, move the pen, and manage canvas properties.

```
public class AppCanvas : IAppCanvas
```

Inheritance

[object](#)  ← AppCanvas

Implements

[IAppCanvas](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

AppCanvas()

Initializes a new instance of the [AppCanvas](#) class. Sets default canvas size and pen color.

```
public AppCanvas()
```

Properties

PenColour

Gets or sets the color of the pen used for drawing.

```
public object PenColour { get; set; }
```

Property Value

[object](#)

Xpos

Gets or sets the current X position of the pen on the canvas.

```
public int Xpos { get; set; }
```

Property Value

[int](#)

Ypos

Gets or sets the current Y position of the pen on the canvas.

```
public int Ypos { get; set; }
```

Property Value

[int](#)

Methods

Circle(Real, bool)

Draws a circle at the current position using the specified radius and fill option.

```
public void Circle(Real radius, bool filled)
```

Parameters

radius [Real](#)

The radius of the circle.

filled [bool](#)

Specifies whether the circle should be filled.

Exceptions

[Exception](#)

Thrown when the radius is negative.

Clear()

Clears the canvas, filling it with a default gray background color.

```
public void Clear()
```

Dispose()

Disposes resources used by the canvas.

```
public void Dispose()
```

DrawTo(int, int)

Draws a line from the current pen position to the specified coordinates.

```
public void DrawTo(int toX, int toY)
```

Parameters

toX [int](#)

The X coordinate to draw to.

toY [int](#)

The Y coordinate to draw to.

Exceptions

[Exception](#)

Thrown when the coordinates are out of bounds.

MoveTo(int, int)

Moves the pen to the specified coordinates without drawing.

```
public void MoveTo(int x, int y)
```

Parameters

x [int](#)

The new X position.

y [int](#)

The new Y position.

Exceptions

[Exception](#)

Thrown when the coordinates are out of bounds.

Rect(int, int, bool)

Draws a rectangle at the current position using the specified dimensions and fill option.

```
public void Rect(int width, int height, bool filled)
```

Parameters

width [int](#)

The width of the rectangle.

height [int](#)

The height of the rectangle.

filled [bool](#)

Specifies whether the rectangle should be filled.

Reset()

Resets the canvas position to the origin (0,0).

```
public void Reset()
```

Set(int, int)

Sets the size of the canvas and initializes the drawing surface.

```
public void Set(int xsize, int ysize)
```

Parameters

xsize [int](#)

The width of the canvas.

ysize [int](#)

The height of the canvas.

SetColour(int, int, int)

Sets the pen color based on the specified RGB values.

```
public void SetColour(int red, int green, int blue)
```

Parameters

red [int](#)

Red component (0-255).

green [int](#)

Green component (0-255).

blue [int](#)

Blue component (0-255).

Exceptions

[Exception](#)

Thrown when any color component is out of range.

SetGraphics(Graphics)

Sets the graphics context for the canvas.

```
public void SetGraphics(Graphics graphics)
```

Parameters

graphics [Graphics](#)

The graphics object to use for drawing.

Tri(int, int)

Draws a triangle at the current position using the specified dimensions.

```
public void Tri(int width, int height)
```

Parameters

width [int](#)

The width of the triangle.

height [int](#)

The height of the triangle.

Exceptions

[Exception](#)

Thrown when the dimensions are invalid.

ValidatePosition(int, int)

Validates whether the specified coordinates are within the canvas bounds.

```
public void ValidatePosition(int x, int y)
```

Parameters

x [int](#)

The X coordinate to validate.

y [int](#)

The Y coordinate to validate.

Exceptions

[Exception](#)

Thrown when the coordinates are out of bounds.

WriteText(string)

Writes text at the current pen position on the canvas.

```
public void WriteText(string text)
```

Parameters

`text` [string](#)

The text to write.

Exceptions

[Exception](#)

Thrown if an error occurs while drawing text.

getBitmap()

Retrieves the bitmap representation of the canvas for rendering.

```
public object getBitmap()
```

Returns

[object](#)

The bitmap object representing the canvas.

Class Form1

Namespace: [ASE Assignment Demo](#)

Assembly: ASE Assignment Demo.dll

Represents the main form of the application where drawing operations can be performed. This form integrates drawing functionality using the BOOSE library, allowing users to execute commands and display graphical output.

```
public class Form1 : Form, IDropTarget, ISynchronizeInvoke, IWin32Window,
    IBindableComponent, IComponent, IDisposable, IContainerControl
```

Inheritance

[object](#) ← [MarshalByRefObject](#) ← [Component](#) ← [Control](#) ← [ScrollableControl](#) ← [ContainerControl](#) ← [Form](#) ← Form1

Implements

[IDropTarget](#), [ISynchronizeInvoke](#), [IWin32Window](#), [IBindableComponent](#), [IComponent](#), [IDisposable](#), [IContainerControl](#)

Inherited Members

[Form.SetVisibleCore\(bool\)](#), [Form.Activate\(\)](#), [Form.ActivateMdiChild\(Form\)](#), [Form.AddOwnedForm\(Form\)](#), [Form.AdjustFormScrollbars\(bool\)](#), [Form.Close\(\)](#), [Form.CreateAccessibilityInstance\(\)](#), [Form.CreateControlsInstance\(\)](#), [Form.CreateHandle\(\)](#), [Form.DefWndProc\(ref Message\)](#), [Form.ProcessMnemonic\(char\)](#), [Form.CenterToParent\(\)](#), [Form.CenterToScreen\(\)](#), [Form.LayoutMdi\(MdiLayout\)](#), [Form.OnActivated\(EventArgs\)](#), [Form.OnBackgroundImageChanged\(EventArgs\)](#), [Form.OnBackgroundImageLayoutChanged\(EventArgs\)](#), [Form.OnClosing\(CancelEventArgs\)](#), [Form.OnClosed\(EventArgs\)](#), [Form.OnFormClosing\(FormClosingEventArgs\)](#), [Form.OnFormClosed\(FormClosedEventArgs\)](#), [Form.OnCreateControl\(\)](#), [Form.OnDeactivate\(EventArgs\)](#), [Form.OnEnabledChanged\(EventArgs\)](#), [Form.OnEnter\(EventArgs\)](#), [Form.OnFontChanged\(EventArgs\)](#), [Form.OnGotFocus\(EventArgs\)](#), [Form.OnHandleCreated\(EventArgs\)](#), [Form.OnHandleDestroyed\(EventArgs\)](#), [Form.OnHelpButtonClicked\(CancelEventArgs\)](#), [Form.OnLayout\(LayoutEventArgs\)](#), [Form.OnLoad\(EventArgs\)](#), [Form.OnMaximizedBoundsChanged\(EventArgs\)](#), [Form.OnMaximumSizeChanged\(EventArgs\)](#), [Form.OnMinimumSizeChanged\(EventArgs\)](#), [Form.OnInputLanguageChanged\(InputLanguageChangedEventArgs\)](#), [Form.OnInputLanguageChanging\(InputLanguageChangingEventArgs\)](#), [Form.OnVisibleChanged\(EventArgs\)](#), [Form.OnMdiChildActivate\(EventArgs\)](#),

[Form.OnMenuStart\(EventArgs\)](#) , [Form.OnMenuComplete\(EventArgs\)](#) ,
[Form.OnPaint\(PaintEventArgs\)](#) , [Form.OnResize\(EventArgs\)](#) ,
[Form.OnDpiChanged\(DpiChangedEventArgs\)](#) , [Form.OnGetDpiScaledSize\(int, int, ref Size\)](#) ,
[Form.OnRightToLeftLayoutChanged\(EventArgs\)](#) , [Form.OnShown\(EventArgs\)](#) ,
[Form.OnTextChanged\(EventArgs\)](#) , [Form.ProcessCmdKey\(ref Message, Keys\)](#) ,
[Form.ProcessDialogKey\(Keys\)](#) , [Form.ProcessDialogChar\(char\)](#) ,
[Form.ProcessKeyPreview\(ref Message\)](#) , [Form.ProcessTabKey\(bool\)](#) ,
[Form.RemoveOwnedForm\(Form\)](#) , [Form.Select\(bool, bool\)](#) ,
[Form.ScaleMinMaxSize\(float, float, bool\)](#) ,
[Form.GetScaledBounds\(Rectangle, SizeF, BoundsSpecified\)](#) ,
[Form.ScaleControl\(SizeF, BoundsSpecified\)](#) ,
[Form.SetBoundsCore\(int, int, int, int, BoundsSpecified\)](#) , [Form.SetClientSizeCore\(int, int\)](#) ,
[Form.SetDesktopBounds\(int, int, int, int\)](#) , [Form.SetDesktopLocation\(int, int\)](#) ,
[Form.Show\(IWin32Window\)](#) , [Form.ShowDialog\(\)](#) , [Form.ShowDialog\(IWin32Window\)](#) ,
[Form.ToString\(\)](#) , [Form.UpdateDefaultButton\(\)](#) , [Form.OnResizeBegin\(EventArgs\)](#) ,
[Form.OnResizeEnd\(EventArgs\)](#) , [Form.OnStyleChanged\(EventArgs\)](#) , [Form.ValidateChildren\(\)](#) ,
[Form.ValidateChildren\(ValidationConstraints\)](#) , [Form.WndProc\(ref Message\)](#) ,
[Form.AcceptButton](#) , [Form.ActiveForm](#) , [Form.ActiveMdiChild](#) , [Form.AllowTransparency](#) ,
[Form.AutoScroll](#) , [Form.AutoSize](#) , [Form.AutoSizeMode](#) , [Form.AutoValidate](#) ,
[Form.BackColor](#) , [Form.FormBorderStyle](#) , [Form.CancelButton](#) , [Form.ClientSize](#) ,
[Form.ControlBox](#) , [Form.CreateParams](#) , [Form.DefaultImeMode](#) , [Form.DefaultSize](#) ,
[Form.DesktopBounds](#) , [Form.DesktopLocation](#) , [Form.DialogResult](#) , [Form.HelpButton](#) ,
[Form.Icon](#) , [Form.IsMdiChild](#) , [Form.IsMdiContainer](#) , [Form.IsRestrictedWindow](#) ,
[Form.KeyPreview](#) , [Form.Location](#) , [Form.MaximizedBounds](#) , [Form.MaximumSize](#) ,
[Form.MainMenuStrip](#) , [Form.MinimumSize](#) , [Form.MaximizeBox](#) , [Form.MdiChildren](#) ,
[Form.MdiChildrenMinimizedAnchorBottom](#) , [Form.MdiParent](#) , [Form.MinimizeBox](#) ,
[Form.Modal](#) , [Form.Opacity](#) , [Form.OwnedForms](#) , [Form.Owner](#) , [Form.RestoreBounds](#) ,
[Form.RightToLeftLayout](#) , [Form.ShowInTaskbar](#) , [Form.ShowIcon](#) ,
[Form.ShowWithoutActivation](#) , [Form.Size](#) , [Form.SizeGripStyle](#) , [Form.StartPosition](#) ,
[Form.Text](#) , [Form.TopLevel](#) , [Form.TopMost](#) , [Form.TransparencyKey](#) , [Form.WindowState](#) ,
[Form.AutoSizeChanged](#) , [Form.AutoValidateChanged](#) , [Form.HelpButtonClicked](#) ,
[Form.MaximizedBoundsChanged](#) , [Form.MaximumSizeChanged](#) ,
[Form.MinimumSizeChanged](#) , [Form.Activated](#) , [Form.Deactivate](#) , [Form.FormClosing](#) ,
[Form.FormClosed](#) , [Form.Load](#) , [Form.MdiChildActivate](#) , [Form.MenuComplete](#) ,
[Form.MenuStart](#) , [Form.InputLanguageChanged](#) , [Form.InputLanguageChanging](#) ,
[Form.RightToLeftLayoutChanged](#) , [Form.Shown](#) , [Form.DpiChanged](#) , [Form.ResizeBegin](#) ,
[Form.ResizeEnd](#) , [ContainerControl.OnAutoValidateChanged\(EventArgs\)](#) ,
[ContainerControl.OnMove\(EventArgs\)](#) , [ContainerControl.OnParentChanged\(EventArgs\)](#) ,
[ContainerControl.PerformAutoScale\(\)](#) , [ContainerControl.RescaleConstantsForDpi\(int, int\)](#) ,
[ContainerControl.Validate\(\)](#) , [ContainerControl.Validate\(bool\)](#) ,

[ContainerControl.AutoScaleDimensions](#) , [ContainerControl.AutoScaleFactor](#) ,
[ContainerControl.AutoScaleMode](#) , [ContainerControl.BindingContext](#) ,
[ContainerControl.CanEnableIme](#) , [ContainerControl.ActiveControl](#) ,
[ContainerControl.CurrentAutoScaleDimensions](#) , [ContainerControl.ParentForm](#) ,
[ScrollableControl.ScrollStateAutoScrolling](#) , [ScrollableControl.ScrollStateHScrollVisible](#) ,
[ScrollableControl.ScrollStateVScrollVisible](#) , [ScrollableControl.ScrollStateUserHasScrolled](#) ,
[ScrollableControl.ScrollStateFullDrag](#) , [ScrollableControl.GetScrollState\(int\)](#) ,
[ScrollableControl.OnMouseWheel\(MouseEventArgs\)](#) ,
[ScrollableControl.OnRightToLeftChanged\(EventArgs\)](#) ,
[ScrollableControl.OnPaintBackground\(PaintEventArgs\)](#) ,
[ScrollableControl.OnPaddingChanged\(EventArgs\)](#) ,
[ScrollableControl.SetDisplayRectLocation\(int, int\)](#) ,
[ScrollableControl.ScrollControlIntoView\(Control\)](#) , [ScrollableControl.ScrollToControl\(Control\)](#) ,
[ScrollableControl.OnScroll\(ScrollEventArgs\)](#) , [ScrollableControl.SetAutoScrollMargin\(int, int\)](#) ,
[ScrollableControl.SetScrollState\(int, bool\)](#) , [ScrollableControl.AutoScrollMargin](#) ,
[ScrollableControl.AutoScrollPosition](#) , [ScrollableControl.AutoScrollMinSize](#) ,
[ScrollableControl.DisplayRectangle](#) , [ScrollableControl.HScroll](#) ,
[ScrollableControl.HorizontalScroll](#) , [ScrollableControl.VScroll](#) , [ScrollableControl.VerticalScroll](#) ,
[ScrollableControl.Scroll](#) , [Control.GetAccessibilityObjectById\(int\)](#) ,
[Control.SetAutoSizeMode\(AutoSizeMode\)](#) , [Control.GetAutoSizeMode\(\)](#) ,
[Control.GetPreferredSize\(Size\)](#) , [Control.AccessibilityNotifyClients\(AccessibleEvents, int\)](#) ,
[Control.AccessibilityNotifyClients\(AccessibleEvents, int, int\)](#) , [Control.BeginInvoke\(Delegate\)](#) ,
[Control.BeginInvoke\(Action\)](#) , [Control.BeginInvoke\(Delegate, params object\[\]\)](#) ,
[Control.BringToFront\(\)](#) , [Control.Contains\(Control\)](#) , [Control.CreateGraphics\(\)](#) ,
[Control.CreateControl\(\)](#) , [Control.DestroyHandle\(\)](#) ,
[Control.DoDragDrop\(object, DragDropEffects\)](#) ,
[Control.DoDragDrop\(object, DragDropEffects, Bitmap, Point, bool\)](#) ,
[Control.DrawToBitmap\(Bitmap, Rectangle\)](#) , [Control.EndInvoke\(IAsyncResult\)](#) ,
[Control.FindForm\(\)](#) , [Control.GetTopLevel\(\)](#) , [Control.RaiseKeyEvent\(object, KeyEventArgs\)](#) ,
[Control.RaiseMouseEvent\(object, MouseEventArgs\)](#) , [Control.Focus\(\)](#) ,
[Control.FromChildHandle\(nint\)](#) , [Control.FromHandle\(nint\)](#) ,
[Control.GetChildAtPoint\(Point, GetChildAtPointSkip\)](#) , [Control.GetChildAtPoint\(Point\)](#) ,
[Control.GetContainerControl\(\)](#) , [Control.GetNextControl\(Control, bool\)](#) ,
[Control.GetStyle\(ControlStyles\)](#) , [Control.Hide\(\)](#) , [Control.InitLayout\(\)](#) ,
[Control.Invalidate\(Region\)](#) , [Control.Invalidate\(Region, bool\)](#) , [Control.Invalidate\(\)](#) ,
[Control.Invalidate\(bool\)](#) , [Control.Invalidate\(Rectangle\)](#) , [Control.Invalidate\(Rectangle, bool\)](#) ,
[Control.Invoke\(Action\)](#) , [Control.Invoke\(Delegate\)](#) ,
[Control.Invoke\(Delegate, params object\[\]\)](#) , [Control.Invoke<T>\(Func<T>\)](#) ,
[Control.InvokePaint\(Control, PaintEventArgs\)](#) ,
[Control.InvokePaintBackground\(Control, PaintEventArgs\)](#) , [Control.IsKeyLocked\(Keys\)](#) ,

[Control.IsInputChar\(char\)](#) , [Control.IsInputKey\(Keys\)](#) , [Control.IsMnemonic\(char, string\)](#) ,
[Control.LogicalToDeviceUnits\(int\)](#) , [Control.LogicalToDeviceUnits\(Size\)](#) ,
[Control.ScaleBitmapLogicalToDevice\(ref Bitmap\)](#) , [Control.NotifyInvalidated\(Rectangle\)](#) ,
[Control.InvokeOnClick\(Control, EventArgs\)](#) , [Control.OnAutoSizeChanged\(EventArgs\)](#) ,
[Control.OnBackColorChanged\(EventArgs\)](#) , [Control.OnBindingContextChanged\(EventArgs\)](#) ,
[Control.OnCausesValidationChanged\(EventArgs\)](#) ,
[Control.OnContextMenuStripChanged\(EventArgs\)](#) , [Control.OnCursorChanged\(EventArgs\)](#) ,
[Control.OnDataContextChanged\(EventArgs\)](#) , [Control.OnDockChanged\(EventArgs\)](#) ,
[Control.OnForeColorChanged\(EventArgs\)](#) , [Control.OnNotifyMessage\(Message\)](#) ,
[Control.OnParentBackColorChanged\(EventArgs\)](#) ,
[Control.OnParentBackgroundImageChanged\(EventArgs\)](#) ,
[Control.OnParentBindingContextChanged\(EventArgs\)](#) ,
[Control.OnParentCursorChanged\(EventArgs\)](#) ,
[Control.OnParentDataContextChanged\(EventArgs\)](#) ,
[Control.OnParentEnabledChanged\(EventArgs\)](#) , [Control.OnParentFontChanged\(EventArgs\)](#) ,
[Control.OnParentForeColorChanged\(EventArgs\)](#) ,
[Control.OnParentRightToLeftChanged\(EventArgs\)](#) ,
[Control.OnParentVisibleChanged\(EventArgs\)](#) , [Control.OnPrint\(PaintEventArgs\)](#) ,
[Control.OnTabIndexChanged\(EventArgs\)](#) , [Control.OnTabStopChanged\(EventArgs\)](#) ,
[Control.OnClick\(EventArgs\)](#) , [Control.OnClientSizeChanged\(EventArgs\)](#) ,
[Control.OnControlAdded\(ControlEventArgs\)](#) , [Control.OnControlRemoved\(ControlEventArgs\)](#) ,
[Control.OnLocationChanged\(EventArgs\)](#) , [Control.OnDoubleClick\(EventArgs\)](#) ,
[Control.OnDragEnter\(DragEventArgs\)](#) , [Control.OnDragOver\(DragEventArgs\)](#) ,
[Control.OnDragLeave\(EventArgs\)](#) , [Control.OnDragDrop\(DragEventArgs\)](#) ,
[Control.OnGiveFeedback\(GiveFeedbackEventArgs\)](#) ,
[Control.InvokeGotFocus\(Control, EventArgs\)](#) , [Control.OnHelpRequested\(HelpEventArgs\)](#) ,
[Control.OnInvalidated\(InvalidateEventArgs\)](#) , [Control.OnKeyDown\(KeyEventArgs\)](#) ,
[Control.OnKeyPress\(KeyPressEventArgs\)](#) , [Control.OnKeyUp\(KeyEventArgs\)](#) ,
[Control.OnLeave\(EventArgs\)](#) , [Control.InvokeLostFocus\(Control, EventArgs\)](#) ,
[Control.OnLostFocus\(EventArgs\)](#) , [Control.OnMarginChanged\(EventArgs\)](#) ,
[Control.OnMouseDoubleClick\(MouseEventArgs\)](#) , [Control.OnMouseClick\(MouseEventArgs\)](#) ,
[Control.OnMouseCaptureChanged\(EventArgs\)](#) , [Control.OnMouseDown\(MouseEventArgs\)](#) ,
[Control.OnMouseEnter\(EventArgs\)](#) , [Control.OnMouseLeave\(EventArgs\)](#) ,
[Control.OnDpiChangedBeforeParent\(EventArgs\)](#) ,
[Control.OnDpiChangedAfterParent\(EventArgs\)](#) , [Control.OnMouseHover\(EventArgs\)](#) ,
[Control.OnMouseMove\(MouseEventArgs\)](#) , [Control.OnMouseUp\(MouseEventArgs\)](#) ,
[Control.OnQueryContinueDrag\(QueryContinueDragEventArgs\)](#) ,
[Control.OnRegionChanged\(EventArgs\)](#) ,
[Control.OnPreviewKeyDown\(PreviewKeyDownEventArgs\)](#) , [Control.OnSizeChanged\(EventArgs\)](#) ,
[Control.OnChangeUICues\(UICuesEventArgs\)](#) , [Control.OnSystemColorsChanged\(EventArgs\)](#) ,

[Control.OnValidating\(CancelEventArgs\)](#), [Control.OnValidated\(EventArgs\)](#),
[Control.PerformLayout\(\)](#), [Control.PerformLayout\(Control, string\)](#),
[Control.PointToClient\(Point\)](#), [Control.PointToScreen\(Point\)](#),
[Control.PreProcessMessage\(ref Message\)](#), [Control.PreProcessControlMessage\(ref Message\)](#),
[Control.ProcessKeyEventArgs\(ref Message\)](#), [Control.ProcessKeyMessage\(ref Message\)](#),
[Control.RaiseDragEvent\(object, DragEventArgs\)](#),
[Control.RaisePaintEvent\(object, PaintEventArgs\)](#), [Control.RecreateHandle\(\)](#),
[Control.RectangleToClient\(Rectangle\)](#), [Control.RectangleToScreen\(Rectangle\)](#),
[Control.ReflectMessage\(nint, ref Message\)](#), [Control.Refresh\(\)](#),
[Control.ResetMouseEventArgs\(\)](#), [Control.ResetText\(\)](#), [Control.ResumeLayout\(\)](#),
[Control.ResumeLayout\(bool\)](#), [Control.Scale\(SizeF\)](#), [Control.Select\(\)](#),
[Control.SelectNextControl\(Control, bool, bool, bool, bool\)](#), [Control.SendToBack\(\)](#),
[Control.SetBounds\(int, int, int, int\)](#), [Control.SetBounds\(int, int, int, int, BoundsSpecified\)](#),
[Control.SizeFromClientSize\(Size\)](#), [Control.SetStyle\(ControlStyles, bool\)](#),
[Control.SetTopLevel\(bool\)](#), [Control.RtlTranslateAlignment\(HorizontalAlignment\)](#),
[Control.RtlTranslateAlignment\(LeftRightAlignment\)](#),
[Control.RtlTranslateAlignment\(ContentAlignment\)](#),
[Control.RtlTranslateHorizontal\(HorizontalAlignment\)](#),
[Control.RtlTranslateLeftRight\(LeftRightAlignment\)](#),
[Control.RtlTranslateContent\(ContentAlignment\)](#), [Control.Show\(\)](#), [Control.SuspendLayout\(\)](#),
[Control.Update\(\)](#), [Control.UpdateBounds\(\)](#), [Control.UpdateBounds\(int, int, int, int\)](#),
[Control.UpdateBounds\(int, int, int, int, int, int\)](#), [Control.UpdateZOrder\(\)](#),
[Control.UpdateStyles\(\)](#), [Control.OnImeModeChanged\(EventArgs\)](#),
[Control.AccessibilityObject](#), [Control.AccessibleDefaultActionDescription](#),
[Control.AccessibleDescription](#), [Control.AccessibleName](#), [Control.AccessibleRole](#),
[Control.AllowDrop](#), [Control.Anchor](#), [Control.AutoScrollOffset](#), [Control.LayoutEngine](#),
[Control.DataContext](#), [Control.BackgroundImage](#), [Control.BackgroundImageLayout](#),
[Control.Bottom](#), [Control.Bounds](#), [Control.CanFocus](#), [Control.CanRaiseEvents](#),
[Control.CanSelect](#), [Control.Capture](#), [Control.CausesValidation](#),
[Control.CheckForIllegalCrossThreadCalls](#), [Control.ClientRectangle](#), [Control.CompanyName](#),
[Control.ContainsFocus](#), [Control.ContextMenuStrip](#), [Control.Controls](#), [Control.Created](#),
[Control.Cursor](#), [Control.DataBindings](#), [Control.DefaultBackColor](#), [Control.DefaultCursor](#),
[Control.DefaultFont](#), [Control.DefaultForeColor](#), [Control.DefaultMargin](#),
[Control.DefaultMaximumSize](#), [Control.DefaultMinimumSize](#), [Control.DefaultPadding](#),
[Control.DeviceDpi](#), [Control.IsDisposed](#), [Control.Disposing](#), [Control.Dock](#),
[Control.DoubleBuffered](#), [Control.Enabled](#), [Control.Focused](#), [Control.Font](#),
[Control.FontHeight](#), [Control.ForeColor](#), [Control.Handle](#), [Control.HasChildren](#),
[Control.Height](#), [Control.IsHandleCreated](#), [Control.InvokeRequired](#), [Control.IsAccessible](#),
[Control.IsAncestorSiteInDesignMode](#), [Control.IsMirrored](#), [Control.Left](#), [Control.Margin](#),
[Control.ModifierKeys](#), [Control.MouseButtons](#), [Control.MousePosition](#), [Control.Name](#),

[Control.Parent](#), [Control.ProductName](#), [Control.ProductVersion](#),
[Control.RecreatingHandle](#), [Control.Region](#), [Control.RenderRightToLeft](#),
[Control.ResizeRedraw](#), [Control.Right](#), [Control.RightToLeft](#), [Control.ScaleChildren](#),
[Control.Site](#), [Control.TabIndex](#), [Control.TabStop](#), [Control.Tag](#), [Control.Top](#),
[Control.TopLevelControl](#), [Control.ShowKeyboardCues](#), [Control.ShowFocusCues](#),
[Control.UseWaitCursor](#), [Control.Visible](#), [Control.Width](#), [Control.PreferredSize](#),
[Control.Padding](#), [Control.ImeMode](#), [Control.ImeModeBase](#),
[Control.PropagatingImeMode](#), [Control.BackColorChanged](#),
[Control.BackgroundImageChanged](#), [Control.BackgroundImageLayoutChanged](#),
[Control.BindingContextChanged](#), [Control.CausesValidationChanged](#),
[Control.ClientSizeChanged](#), [Control.ContextMenuStripChanged](#), [Control.CursorChanged](#),
[Control.DockChanged](#), [Control.EnabledChanged](#), [Control.FontChanged](#),
[Control.ForeColorChanged](#), [Control.LocationChanged](#), [Control.MarginChanged](#),
[Control.RegionChanged](#), [Control.RightToLeftChanged](#), [Control.SizeChanged](#),
[Control.TabIndexChanged](#), [Control.TabStopChanged](#), [Control.TextChanged](#),
[Control.VisibleChanged](#), [Control.Click](#), [Control.ControlAdded](#), [Control.ControlRemoved](#),
[Control.DataContextChanged](#), [Control.DragDrop](#), [Control.DragEnter](#), [Control.DragOver](#),
[Control.DragLeave](#), [Control.GiveFeedback](#), [Control.HandleCreated](#),
[Control.HandleDestroyed](#), [Control.HelpRequested](#), [Control.Invalidated](#),
[Control.PaddingChanged](#), [Control.Paint](#), [Control.QueryContinueDrag](#),
[Control.QueryAccessibilityHelp](#), [Control.DoubleClick](#), [Control.Enter](#), [Control.GotFocus](#),
[Control.KeyDown](#), [Control.KeyPress](#), [Control.KeyUp](#), [Control.Layout](#), [Control.Leave](#),
[Control.LostFocus](#), [Control.MouseClick](#), [Control.MouseDoubleClick](#),
[Control.MouseCaptureChanged](#), [Control.MouseDown](#), [Control.MouseEnter](#),
[Control.MouseLeave](#), [Control.DpiChangedBeforeParent](#), [Control.DpiChangedAfterParent](#),
[Control.MouseHover](#), [Control.MouseMove](#), [Control.MouseUp](#), [Control.MouseWheel](#),
[Control.Move](#), [Control.PreviewKeyDown](#), [Control.Resize](#), [Control.ChangeUICues](#),
[Control.StyleChanged](#), [Control.SystemColorsChanged](#), [Control.Validating](#),
[Control.Validated](#), [Control.ParentChanged](#), [Control.ImeModeChanged](#),
[Component.Dispose\(\)](#), [Component.GetService\(Type\)](#), [Component.Container](#),
[Component.DesignMode](#), [Component.Events](#), [Component.Disposed](#),
[MarshalByRefObject.GetLifetimeService\(\)](#), [MarshalByRefObject.InitializeLifetimeService\(\)](#),
[MarshalByRefObject.MemberwiseClone\(bool\)](#), [object.Equals\(object\)](#),
[object.Equals\(object, object\)](#), [object.GetHashCode\(\)](#), [object.GetType\(\)](#),
[object.MemberwiseClone\(\)](#), [object.ReferenceEquals\(object, object\)](#)

Constructors

Form1()

Initializes a new instance of the [Form1](#) class. Sets up the canvas and initializes the drawing context, including the configuration of BOOSE library components such as the parser and program storage.

```
public Form1()
```

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

disposing [bool](#)

true if managed resources should be disposed; otherwise, false.

Interface IAppCanvas

Namespace: [ASE Assignment Demo](#)

Assembly: ASE Assignment Demo.dll

Extends the base interface with additional functionalities for canvas operations.

```
public interface IAppCanvas
```

Methods

Circle(Real, bool)

Draws a circle at the current position on the canvas.

```
void Circle(Real radius, bool filled)
```

Parameters

radius [Real](#)

The radius of the circle.

filled [bool](#) 

Whether the circle should be filled.

MoveTo(int, int)

Moves the drawing position to the specified coordinates on the canvas.

```
void MoveTo(int x, int y)
```

Parameters

x [int](#) 

The X coordinate to move to.

y [int](#)

The Y coordinate to move to.

Rect(int, int, bool)

Draws a rectangle at the current position on the canvas.

```
void Rect(int width, int height, bool filled)
```

Parameters

width [int](#)

The width of the rectangle.

height [int](#)

The height of the rectangle.

filled [bool](#)

Whether the rectangle should be filled (true) or just outlined (false).

Reset()

Resets the canvas position to (0, 0).

```
void Reset()
```

SetColour(int, int, int)

Sets the pen color using RGB values.

```
void SetColour(int red, int green, int blue)
```

Parameters

red [int](#)

Red component (0-255).

green [int](#)

Green component (0-255).

blue [int](#)

Blue component (0-255).

SetGraphics(Graphics)

Sets the graphics context for the canvas.

```
void SetGraphics(Graphics graphics)
```

Parameters

graphics [Graphics](#)

The graphics object used for rendering.

Tri(int, int)

Draws a triangle at the current position on the canvas.

```
void Tri(int width, int height)
```

Parameters

width [int](#)

The width of the triangle.

height [int](#)

The height of the triangle.

ValidatePosition(int, int)

Validates whether the given position is within the bounds of the canvas.

```
void ValidatePosition(int x, int y)
```

Parameters

x [int](#)

The X coordinate of the position to validate.

y [int](#)

The Y coordinate of the position to validate.

WriteText(string)

Writes text at the current position on the canvas.

```
void WriteText(string text)
```

Parameters

text [string](#)

The text to be written.

getBitmap()

Retrieves the current bitmap of the canvas.

```
object getBitmap()
```

Returns

[object](#) 

The bitmap object representing the current state of the canvas.

Namespace ASE_Assignment_Demo.

Commands

Classes

[CircleCommand](#)

Command that represents drawing a circle on the canvas.

[MoveToCommand](#)

Command that represents moving the drawing position on the canvas.

[RectCommand](#)

Command that represents drawing a rectangle on the canvas.

[SetPenCommand](#)

Command that represents setting the pen color on the canvas.

[TriangleCommand](#)

Represents a command to draw a triangle on the canvas.

[WhileCommand](#)

Represents a command that repeatedly executes a set of commands while a specified condition is true.

Class CircleCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Command that represents drawing a circle on the canvas.

```
public class CircleCommand : ICustomCommand
```

Inheritance

[object](#)  ← CircleCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

CircleCommand(Real, bool)

Initializes a new instance of the [CircleCommand](#) class.

```
public CircleCommand(Real radius, bool filled)
```

Parameters

radius [Real](#)

The radius of the circle.

filled [bool](#) 

Indicates whether the circle should be filled.

Exceptions

[ArgumentNullException](#) 

Thrown if `radius` is null.

Methods

Execute(IAppCanvas)

Executes the circle drawing command on the given canvas.

```
public void Execute(IAppCanvas canvas)
```

Parameters

`canvas` [IAppCanvas](#)

The canvas on which to draw the circle.

Exceptions

[ArgumentNullException](#) 

Thrown if `canvas` is null.

Class MoveToCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Command that represents moving the drawing position on the canvas.

```
public class MoveToCommand : ICustomCommand
```

Inheritance

[object](#)  ← MoveToCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

MoveToCommand(Int, Int)

Initializes a new instance of the [MoveToCommand](#) class.

```
public MoveToCommand(Int x, Int y)
```

Parameters

x [Int](#)

The X coordinate to move to.

y [Int](#)

The Y coordinate to move to.

Methods

Execute(IAppCanvas)

Executes the command to move the drawing position on the canvas.

```
public void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas on which to move the drawing position.

Class RectCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Command that represents drawing a rectangle on the canvas.

```
public class RectCommand : ICustomCommand
```

Inheritance

[object](#)  ← RectCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  , [object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  , [object.ToString\(\)](#) 

Constructors

RectCommand(Int, Int, bool)

Initializes a new instance of the [RectCommand](#) class.

```
public RectCommand(Int width, Int height, bool filled)
```

Parameters

width [Int](#)

The width of the rectangle.

height [Int](#)

The height of the rectangle.

filled [bool](#) 

Indicates whether the rectangle should be filled (true) or outlined (false).

Methods

Execute(IAppCanvas)

Executes the command to draw a rectangle on the canvas.

```
public void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas on which to draw the rectangle.

Class SetPenCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Command that represents setting the pen color on the canvas.

```
public class SetPenCommand : ICustomCommand
```

Inheritance

[object](#)  ← SetPenCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

SetPenCommand(Int, Int, Int)

Initializes a new instance of the [SetPenCommand](#) class.

```
public SetPenCommand(Int red, Int green, Int blue)
```

Parameters

red [Int](#)

The red component of the pen color (0-255).

green [Int](#)

The green component of the pen color (0-255).

blue [Int](#)

The blue component of the pen color (0-255).

Methods

Execute(IAppCanvas)

Executes the command to set the pen color on the canvas.

```
public void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas to apply the pen color.

Class TriangleCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Represents a command to draw a triangle on the canvas.

```
public class TriangleCommand : ICustomCommand
```

Inheritance

[object](#)  ← TriangleCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

TriangleCommand(Int, Int)

Initializes a new instance of the [TriangleCommand](#) class.

```
public TriangleCommand(Int width, Int height)
```

Parameters

width [Int](#)

The width of the triangle.

height [Int](#)

The height of the triangle.

Methods

Execute(IAppCanvas)

Executes the triangle drawing command on the canvas.

```
public void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas on which to draw the triangle.

Class WhileCommand

Namespace: [ASE Assignment Demo.Commands](#)

Assembly: ASE Assignment Demo.dll

Represents a command that repeatedly executes a set of commands while a specified condition is true.

```
public class WhileCommand : ICustomCommand
```

Inheritance

[object](#)  ← WhileCommand

Implements

[ICustomCommand](#)

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Constructors

WhileCommand(string, Dictionary<string, object>, CommandFactory)

Initializes a new instance of the [WhileCommand](#) class.

```
public WhileCommand(string condition, Dictionary<string, object> variables,  
CommandFactory factory)
```

Parameters

condition [string](#) 

The condition that determines whether the loop continues executing.

variables [Dictionary](#)  <[string](#) , [object](#)  >

A dictionary containing variables used within the condition and loop.

factory [CommandFactory](#)

A factory for creating commands dynamically within the loop.

Methods

AddCommand(ICustomCommand)

Adds a command to the list of commands to be executed in each iteration of the loop.

```
public void AddCommand(ICustomCommand command)
```

Parameters

command [ICustomCommand](#)

The command to add to the loop.

Execute(IAppCanvas)

Executes the while loop, evaluating the condition and executing the loop commands while the condition evaluates to true.

```
public void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas on which the commands are executed.

Exceptions

[ArgumentException](#) 

Thrown if the condition is invalid or contains unknown operators.

Namespace ASE_Assignment_Demo.Dynamic Classes

[Int](#)

Represents an integer value encapsulated within a custom class.

[Real](#)

Represents a wrapper for real (double) values.

Class Int

Namespace: [ASE Assignment Demo.Dynamic](#)

Assembly: ASE Assignment Demo.dll

Represents an integer value encapsulated within a custom class.

```
public class Int
```

Inheritance

[object](#)  ← Int

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#) 

Constructors

Int(int)

Initializes a new instance of the [Int](#) class with a specified value.

```
public Int(int value)
```

Parameters

value [int](#) 

The integer value to encapsulate.

Methods

GetValue()

Retrieves the encapsulated integer value.

```
public int GetValue()
```

Returns

[int](#)

The integer value.

SetValue(int)

Sets a new value for the encapsulated integer.

```
public void SetValue(int value)
```

Parameters

value [int](#)

The new integer value to set.

ToString()

Returns a string representation of the encapsulated integer value.

```
public override string ToString()
```

Returns

[string](#)

The string representation of the integer value.

Class Real

Namespace: [ASE Assignment Demo.Dynamic](#)

Assembly: ASE Assignment Demo.dll

Represents a wrapper for real (double) values.

```
public class Real
```

Inheritance

[object](#)  ← Real

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#) 

Constructors

Real(double)

Initializes a new instance of the [Real](#) class with a specified real value.

```
public Real(double value)
```

Parameters

value [double](#) 

The real value to encapsulate.

Methods

GetValue()

Retrieves the stored real value.

```
public double GetValue()
```

Returns

[double](#)[↗]

The stored real value.

SetValue(double)

Sets a new value for the stored real value.

```
public void SetValue(double value)
```

Parameters

value [double](#)[↗]

The new real value to set.

ToString()

Returns a string representation of the stored real value.

```
public override string ToString()
```

Returns

[string](#)[↗]

The string representation of the real value.

Namespace ASE_Assignment_Demo.Factories

Classes

[CommandFactory](#)

Factory class responsible for creating instances of specific commands based on the provided command name.

Class CommandFactory

Namespace: [ASE Assignment Demo.Factories](#)

Assembly: ASE Assignment Demo.dll

Factory class responsible for creating instances of specific commands based on the provided command name.

```
public class CommandFactory
```

Inheritance

[object](#)  ← CommandFactory

Inherited Members

[object.Equals\(object\)](#)  , [object.Equals\(object, object\)](#)  , [object.GetHashCode\(\)](#)  ,
[object.GetType\(\)](#)  , [object.MemberwiseClone\(\)](#)  , [object.ReferenceEquals\(object, object\)](#)  ,
[object.ToString\(\)](#) 

Methods

CreateCommand(string, object[], List<ICustomCommand>)

Creates an instance of a specific command based on the provided command name and arguments.

```
public ICustomCommand CreateCommand(string commandName, object[] args,  
List<ICustomCommand> bodyCommands = null)
```

Parameters

commandName [string](#) 

The name of the command to create.

args [object](#)  []

The arguments to pass to the command constructor.

bodyCommands [List](#)  <[ICustomCommand](#)>

Optional. Commands for block structures like while or if.

Returns

[ICustomCommand](#)

An instance of the corresponding [ICustomCommand](#).

Exceptions

[ArgumentException](#) 

Thrown if the command name is unknown or if the arguments are invalid for the command.

Namespace ASE_Assignment_Demo.

Interfaces

Interfaces

[ICustomCommand](#)

Represents a command that can be executed on a canvas.

Interface ICustomCommand

Namespace: [ASE Assignment Demo.Interfaces](#)

Assembly: ASE Assignment Demo.dll

Represents a command that can be executed on a canvas.

```
public interface ICustomCommand
```

Methods

Execute(IAppCanvas)

Executes the command on the specified canvas.

```
void Execute(IAppCanvas canvas)
```

Parameters

canvas [IAppCanvas](#)

The canvas on which the command is executed.